

Exercises 5.10

Exercise 5.10.1

Recall that $\neg x$ is the negative of the Boolean variable x .

- Show that a single perceptron can learn the Boolean function $y = x_1 \wedge \neg x_2$, with some $x_1, x_2 \in \{0, 1\}$.
- The same question as in part *a* for the Boolean function $y = x_1 \vee \neg x_2$, with some $x_1, x_2 \in \{0, 1\}$.
- Show that a perceptron with one Boolean input, x , can learn the negation function $y = \neg x$. What about the linear neuron?
- Show that a perceptron with three Boolean inputs, x_1, x_2, x_3 , can learn the negation function $y = \neg x$. What about $x_1 \vee x_2 \vee x_3$?

Exercise 5.10.2

Show that two finite linearly separable sets A and B can be separated by a perceptron with rational weights.

Exercise 5.10.3

Assume the inputs to a linear neuron are independent and normally distributed, $X_i \sim \mathcal{N}(0, \sigma_i^2)$, $i = 1, \dots, n$. Find the optimal weights, w^*

- A one dimensional random variable with zero mean, Z is learned by a linear neuron with input X . Assume the input, X , and the target Z are independent. Write the cost function and find the optimal parameters, w^* . Provide an interpretation of the result.
- Use Newton's method to obtain the optimal parameters of a linear neuron.

Exercise 5.10.4

Explain the equivalence between the linear regression algorithm and the learning of a linear neuron.

Exercise 5.10.5

Consider a neuron with a continuum input, whose output is $y = H(\int_0^1 x d\mu(x))$. Find the output in the case when the measure is $\mu = \delta_{x_0}$.

Exercise 5.10.6

Consider n points $P_1, \dots, P_n$, included in a half-circle, and denote by $\mathbf{x}_1, \dots, \mathbf{x}_n$ their coordinate vectors.

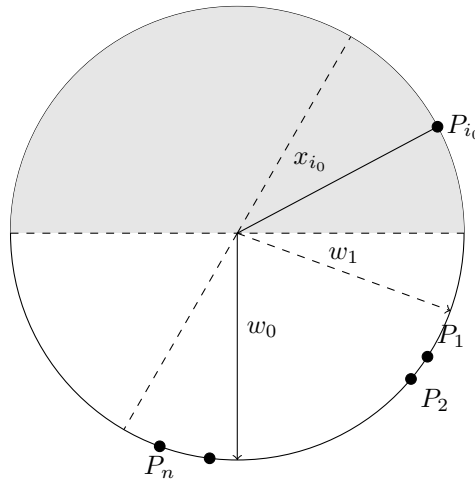


Figure 1: Perceptron algorithm

A perceptron can learn the aforementioned half-circle by the following algorithm:

1. Start from an arbitrary half-circle determined by its diameter and a unit normal vector w_0 . Then select an incorrectly classified point, P_{i_0} , i.e., a point for which $\langle w_0, \mathbf{x}_{i_0} \rangle < 0$. See Fig 1.
2. Rotate the diameter that the new normal is $w_1 = w_0 + \mathbf{x}_{i_0}$. Show that the point is now correctly classified.
3. Repeating the previous two steps, we constructed inductively the sequence of vectors $(w_m)_m$. Such that $w_{m+1} = w_m + \mathbf{x}_{i_m}$, where P_{i_m} is a point misclassified at step m . Show that the process ends in a finite number of steps, i.e., there is a $N > 1$ such that $\langle w_0, \mathbf{x}_j \rangle > 0$, $\forall 1 \leq j \leq n$. Find an estimate of the number N .

Exercise 5.10.7

Modify the perceptron learning algorithm given by Exercise 5.10.6 for the case when the points $P_1, \dots, P_n$ are included in a half-plane.

Exercise 5.10.8

Let $\mathbf{1}_A(x)$ denote the characteristic function of the set A , namely, $\mathbf{1}_A(x) = 1$ if $x \in A$ and $\mathbf{1}_A(x) = 0$ if $x \notin A$

- a. Show that the function $\phi(x_1, x_2) = \mathbf{1}_{x_2 > x_1 + 0.5}(x_1, x_2) + \mathbf{1}_{x_2 > x_1 - 0.5}(x_1, x_2)$ implements *XOR*.
- b. Show that the *XOR* function can be implemented by a linear combination of the two perceptrons.

Exercise 5.10.9

Show that if all input vectors x_k have the same length, then the α -LMS algorithm minimizes the mean square error and in this case the updating rule (5.6.6) becomes the gradient descent rule.

Exercise 5.10.10

Find the weights of a Madaline with two Adaline units which implements the *XNOR* function.

SOLUTIONS

Exercise 5.10.1 (a)

Let's analyze the values taken by the boolean function $f(x_1, x_2) = x_1 \wedge \neg x_2$. We get the following table:

x_1	x_2	$\neg x_2$	$x_1 \wedge \neg x_2$
0	0	1	0
0	1	0	0
1	0	1	1
1	1	0	0

It's easy to see that the function $A(x_1, x_2) := \varphi((1, -1) \cdot (x_1, x_2) - \frac{1}{2}) = \begin{cases} 1 & \text{if } x_1 - x_2 \geq \frac{1}{2} \\ 0 & \text{if } x_1 - x_2 < \frac{1}{2} \end{cases}$ is an adequate choice because it can learn the points presented in the above. This in turn means that the desired neuron is given by the quadruple $((-1, x_1, x_2), (1, -1, \frac{1}{2}), A, (0, 0, 1, 0))$.

Exercise 5.10.1 (b)

Let's analyze the values taken by the boolean function $f(x_1, x_2) = x_1 \wedge \neg x_2$. We get the following table:

x_1	x_2	$\neg x_2$	$x_1 \vee \neg x_2$
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1

It's easy to see that the function $\varphi(x_1, x_2) =$ is an adequate choice because it can learn the points presented in the above. This in turn means that $(\{0, 1\}^2, \mathbf{I}_{\{x_1+x_2>0\}}(x_1, x_2), \{0, 1\}^2)$

Exercise 5.10.1 (c)

Let's analyze the values taken by the boolean function $f(x_1) = \neg x_1$. We get the following table:

x_1	$\neg x_1$
0	1
1	0

It's easy to see that the function $\mathbf{I}_{\{x_1+x_2>0\}}(x_1, x_2)$ is an adequate choice because it can learn the points presented in the above. This in turn means that $(\{0, 1\}^2, \mathbf{I}_{\{x_1+x_2>0\}}(x_1, x_2), \{0, 1\}^2)$

Exercise 5.10.1 (d)

Let's analyze the values taken by the boolean function $f(x_1, x_2) = x_1 \wedge \neg x_2$. We get the following table:

x_1	x_2	$\neg x_2$	$x_1 \vee \neg x_2$
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1

On the other hand, analyzing the boolean function ...

x_1	x_2	$\neg x_2$	$x_1 \vee \neg x_2$
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1

It's easy to see that the function $\mathbf{I}_{\{x_1+x_2>0\}}(x_1, x_2)$ is an adequate choice because it can learn the points presented in the above. This in turn means that $(\{0, 1\}^2, \mathbf{I}_{\{x_1+x_2>0\}}(x_1, x_2), \{0, 1\}^2)$

Exercise 5.10.2